PROFESSIONAL EDUCATION



SUMMER 2024

ADVANCED REINFORCEMENT LEARNING

DATES	DURATION	FORMAT	FEE	CEUS
August 1–2, 2024	2 Days	On Campus or Live Virtual	$2,500	1.6

Explore the cutting-edge of RL research

Reinforcement learning (RL) is transforming machine learning applications across industries—and its potential is only beginning to be tapped. From natural language processing and computer vision to self-driving cars and gaming, this paradigm offers practical applications in industries as diverse as transportation, retail, finance, urban planning, and healthcare.

In this accelerated, two-day course, you'll receive an advanced overview of the cutting-edge RL topics that are driving exciting advancements in machine learning. Through interactive lectures and exercises, you'll acquire a multi-faceted glimpse into the development and potential of RL, from the perspectives of statistics, optimal control, economics, operational research, and other disciplines.

You will additionally have the opportunity to put your learning into practice during hands-on clinics, in which you will use advanced algorithms to solve real-world problems, and then discuss your solutions with the class and instructors during office hours. You will leave the course armed with a broad understanding of reinforcement learning as a tool, mathematical framework, and an active field of study.

EARN A PROFESSIONAL CERTIFICATE

This course may be taken individually or as part of the ***Professional Certificate Program in Machine Learning & Artificial Intelligence.***

WHO SHOULD ATTEND

This course is designed for mid-career professionals who are actively involved in or want to learn more about reinforcement learning. The strategies covered will be applicable for a wide variety of fields, including robotics, automotive, manufacturing, urban planning and design, logistics, government and military, science and technology, retail, finance, healthcare, and pharmaceutical industries. The curriculum will be particularly helpful for:

- **Research scientists** looking to enhance their ability to leverage advanced deep RL tools, such as multi-agent RL, offline RL, learning from demonstrations, and curriculum learning
- **Data scientists** who are already familiar with RL strategies, but want to take their machine learning toolkit to the next level
- **Data analysts and business analysts** who are responsible for solving challenges that require extracting insights from large sets of unstructured data
- **Machine learning engineers and software engineers** who want to use advanced RL models to overcome practical constraints in existing projects
- **Product managers and program managers** who need to discern when it's appropriate and effective to apply RL
- **CTOs and other technology leaders** who want a deeper understanding of how advanced RL strategies can be harnessed to help address and overcome their organizational roadblocks

professional.mit.edu/arl

LEARNING OUTCOMES

Each problem is unique: the tradeoffs are unique, the challenges are different. Therefore, simply applying Deep RL algorithms will not work. For instance, you might have an existing dataset that can be leveraged or there is already a working system that only needs to be improved or maybe the problem has a long-time horizon, or that the environment is constantly changing. First what RL framework to use? Is it offline reinforcement learning, model-based RL, curriculum learning, hierarchical RL, learning from demonstrations, multi-agent RL? Second, how do these algorithms work and how to use them. This course will address these questions and practical challenges in using Deep RL algorithms. By the end of the course you will have rules of thumb to decide which RL framework to use and how to implement it.

By completing this course, you will enhance your ability to:

- Improve the performance of supervised learning systems by fine-tuning with RL methods
- Decide when to use an analytical model or when to use a learned model. Learn about Model-Based RL
- Learn techniques for applying Deep RL methods to practical problems when it is impossible to collect large amounts of data
- Determine the reinforcement learning framework (e.g. goal-directed, hierarchical, offline reinforcement learning, bandits) that is best-suited to solve a specific problem
- Learn about RL algorithms that can generalize to new goals
- Learn more about what to do if the reward is very sparse or the time horizon is very long
- Select the most promising algorithms for an already-formulated reinforcement learning problem
- Recognize the limitations of reinforcement learning in order to judge whether a situation is suited for these strategies

LEAD INSTRUCTORS

Pulkit Agrawal



Pulkit Agrawal is the Steven and Renee Chair Assistant Professor of Electrical Engineering and Computer Science at MIT and leads the Improbable AI Lab, part of the Computer Science and Artificial Intelligence Lab at MIT and affiliated with the Laboratory for Information and Decision Systems. Dr. Agrawal co-founded SafelyYou, an organization that builds fall prevention technology, and the AI Foundry, an incubator for AI startups. He currently serves as an advisor for several startups and has research collaborations with companies such as IBM, Toyota, Sony, and Facebook AI Research (FAIR).

Cathy Wu



Cathy Wu is the Gilbert W. Winslow Career Development Assistant Professor of civil and environmental engineering at MIT and has worked across many fields and organizations, including Microsoft Research, OpenAI, the Google X Self-Driving Car Team, AT&T, Caltrans, Facebook, and Dropbox. Wu is also the founder and Chair of the Interdisciplinary Research Initiative at the ACM Future of Computing Academy.

Teaching Methodology

The certificate curriculum is grounded in the spirit of MIT's motto, "Mens et Manus," or "mind and hand," which combines theoretical instruction with hands-on methods of discovery.



INTERACTIVE LECTURES



HANDS-ON EXERCISES



GROUP PROJECTS



SUPPLEMENTAL RESOURCES

PROFESSIONAL EDUCATION

ABOUT MIT PROFESSIONAL EDUCATION

For 70 years, MIT Professional Education has been providing technical professionals worldwide a gateway to renowned MIT research, knowledge, and expertise, through advanced education programs designed specifically for them. In addition to industry-focused, two-to-five-day live virtual and on-campus courses through Short Programs, MIT Professional Education offers professionals the opportunity to take online and blended learning courses through Digital Plus Programs, attend courses abroad through International Programs, enroll in regular MIT academic courses through the Advanced Study Program, or attend Custom Programs designed specifically for their companies. For more information, please visit **professional.mit.edu**.